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Week 1

Project Plan 12th June.

The main focus will be increasing the spatial resolution of wildfire satellite images. Step 1 of the big project. If we have time we can go to step 2.

Paper to follow:

https://www.mdpi.com/2072-4292/16/4/715

https://www.sciencedirect.com/science/article/abs/pii/S092427162100160X?via%3Dihub
(30m spatial resolution)

1. GOES vs **MTG**. If it's in Google earth engine it's easier to collect data. What bands are we using? Etc
2. Sentinel-2(20m) vs Landsat(30m) what do we want?
3. Cascade approach or direct approach, or both and compare

Coarse Input Data: GOES vs MTG

	GOES	**MTG**
Spatial Res	2km	Visible and NIR: 1km IR: 2km (Active Fire Monitoring EO:EUM:DAT:0682)
Temporal Res	10 mins	Every 10 mins
Bands	3.9μm	FCI channel IR-3.8μm
Spatial coverage	The Americas (North and South)	Lat: -67.5° to 67.5°, Lon: -67.5° to 67.5° (EU, Africa, Atlantic Ocean, parts of Middle East and South America?)

Temporal extent	GOES-16: May 2017 - Now GOES-17: 2018 - now GOES-18: Oct 2022 - Now	Jul 2025 - Now
Access	GEE API	EUMETSAT API
Advantages	Can access via GEE API	Directly relevant to the ESA project. Greater spatial extent
Limitations	Only covers the Americas	Data only available from July 2025
Relevant links	GEE GOES-16 FDCF Data Guide	Data guide

- Since GOES likely will have more wildfire data (or could check if the spatial breadth of the MTG compensates for the short time period), we could initially work with GOES as our coarse input data. If time permits, we can try tuning the model with MTG?
- Alternatively we can combine MTG and MSG data to get a larger dataset for MTG (provided that the relevant data have the same spatial resolution. The IR bands of MSG are 3km while 1/2km for MTG.
- For MTG, we find that some bands have different resolution (0.5, 1, 2km) due to the different modes (High Resolution Fast Imagery (HRFI))
- Fire Radiative Power - MTG have data available in 1km from May 2026

High-res Target Data: Landsat vs Sentinel-2

	Landsat	**Sentinel-2**
Spatial res	30m	20m
Temporal res	8/16 days	5 days
Relevant Bands	Band 6 (1.6 μm), Band 7 (2.2 μm)	Band 11 (1.6 μm), Band 12 (2.2 μm)
Temporal extent	Landat 8: Apr 2013 - present Landsat 9: Sept 2019 - present In NASA FIRMS: June	March 2017 - present

	2022 - present	
Spatial extent	NASA FIRMS: US/Canada only	global
Access	GEE	Copernicus Data Space Ecosystem (API)
Advantages	Contains thermal infrared band Fire dataset is available	Slightly higher resolution
Limitations	NASA FIRMS only have data from 2022, and only available for US/Canada	Lacks thermal infrared bands. As far as I can find, there is no clear 'fire' dataset that I could use

- Active fire detection in Landsat-8 imagery: A large-scale dataset and a deep-learning study Only fire data for Aug/Sept 2020 (worldwide).

Intermediate Resolution Data

- VIIRS:
  - Resolution: 375m
  - Access via NASA FIRMS
  - Temporal Res: Daily
- Main problem: when doing the cascaded approach, we need to spatially and temporally match VIIRS with the high-res target (landsat or Sentinel). There could potentially be a temporal mismatch

Summary: Likely use GOES for input coarse data, VIIRS for intermediate, and Landsat (Pereira et al. dataset) for 'ground truth'

Main Research

I have chosen to compare the following architectures:

Experiment	Architecture	Pipeline Structure	Input Resolution	Output resolution	Model type
E1	EDiffSR	Direct	2km (GOES)	30m (Landsat, Pereira et al.)	Diffusion

E2	EDiffSR	Cascaded	Stage 1: 2km -> 375m Stage 2: 375m -> 30m	30m (Landsat, Pereira et al.)	Diffusion
E3	HAUNet	Direct	2km (GOES)	30m (Landsat, Pereira et al.)	Attention UNet
E4	HAUNet	Cascaded	Stage 1: 2km -> 375m Stage 2: 375m -> 30m	30m (Landsat, Pereira et al.)	Attention UNet
E5	ESTNet	Direct	2km (GOES)	30m (Landsat, Pereira et al.)	Transformer
E6	ESTNet	Cascaded	Stage 1: 2km -> 375m Stage 2: 375m -> 30m	30m (Landsat, Pereira et al.)	Transformer

So for the Cascade approach, first you train a model that goes from 2km to 375m. Then we use that model to increase the res of Goes so we can match Goes and Landsat.

WEEK 2

Objectives:

1. Only satellites or more data? (Landcover, DEM, …). Please select the bands we are using in the satellites.
2. Evaluation metric (what we use to know if the result is good, MSE? Something that measures the blurriness)
3. Decide 3 architectures to try.
4. Ask gemini for a plan to train the model in Google Cloud Platform. So ask gemini what is the best way to ingest the data (data pipeline big query ? or?). IRP plan

Inputs/outputs/possible covariates:

- **GOES-16/17/18** bands (from https://www.sciencedirect.com/science/article/pii/S2666017221000031):
    - Active fire bands: 3.80–3.99 µm and 10.82–11.60 µm at 2km
    - Alternatively, use Level 2 Fire Mask (FSH) or Fire Radiative Power (FRP) bands directly, at 2 km.
- **VIIRS** (375m):
    - Use NASA FIRMS data, which uses I-bands: Bright_ti4, Bright_ti5, frp (fire radiative power) for thermal anomaly detection.

- **Topography:** Copernicus DEM GLO-30 (30m)
- **Landcover:** ESA WorldCover 10m v100 **(10m) (Here it tells you the water bodies)**
- **NDWI (Normalised Difference Water Index):** Landsat Collection 2 Tier 1 Level 2 32-Day NDWI Composite **(30m) (dont need this)**
- **NDVI (Normalised Difference Vegetation Index):** Landsat Collection 2 Tier 1 Level 2 8-Day NDVI Composite **(30m)**
- **Global inland water: GLCF:** Landsat Global Inland Water **(30m)**
- **Buildings: GHSL:** Global settlement characteristics (10 m) 2018 (P2023A) **(10m)**
- **Landsat** (Pereira et al. dataset) (30m)
    - Uses 10 bands: Channels b1 to b11 except b8: Green, Red, Near-Infrared (NIR), Shortwave Infrared 1 (SWIR 1), Shortwave Infrared 2 (SWIR 2), Cirrus, Thermal Infrared 1 (TIRS 1), Thermal Infrared 2 (TIRS 2)

Metrics

- IOU/F1 (binary mask):
    - IOU Quantifies the overlap between the predicted fire mask and the ground truth mask.
    - F1: The harmonic mean of precision (how many predicted fires were real) and recall (how many real fires were detected).
- (PSNR/SSIM: standard image-based metrics)
- FID: evaluates the quality of generative outputs by comparing the statistical distribution of generated images to real images.
- MSE/RMSE
- CG-FG MSE (Coarse-Grid to Fine-Grid Mean Squared Error) and SR-FG MSE (Super-Resolution to Fine-Grid Mean Squared Error)

Data Processing Pipeline

From "Deep Learning Approach to Improve Spatial Resolution of GOES-17 Wildfire Boundaries Using VIIRS Satellite Data":

- VIIRS (VNP14IMGTDL_NRT) in vector CSV format, GOES in raster NetCDF format.
- GOES contain fire and background info, while VIIRS contain the location and radiance value of globally detected fire hotspots.
- For GOES, author used channel 7 (IR shortwave) of Level 1B (L1B) Radiances product (ABI-L1B-Rad) as input.
- Used a list of wildfire events in the western U.S. between 2019 and 2021 (the link no longer works) to obtain ROI
- Preprocessing:
    1. Extract fire data from VIIRS identifying the timestamps. First extracts the records from the VIIRS CSV file to identify detected fire hotspots that fall within the ROI and duration of wildfire event. The pipeline also identifies unique timestamps from the extracted records.

2. Download GOES images for each identified timestamp. The pre-processing pipeline downloads GOES images with captured times that are near to each VIIRS timestamp identified in Step 1. GOES have a temporal resolution of 5 min, meaning that there will always be a GOES image within 2.5 min of the VIIRS captured time, except in cases where the GOES file is corrupted due to cooling system issues. In case of corrupted GOES data, Steps 3 and 4 will be halted and the pipeline will proceed to the next timestamp.
3. Creating processed GOES images. The GOES images obtained in Step 2 have different projections from the corresponding VIIRS. The GOES images are cropped to match the site's ROI and reprojected into a standard coordinate reference system (CRS).
4. Creating processed VIIRS images. The VIIRS records in Step 1 are grouped by timestamp and rasterized, interpolated, and saved into GeoTIFF images using the same projection as the one used to reproject GOES images in Step 3.

- **GOES Processing:**
    - The author virtually mounted NOAA's public AWS S3 storage bucket (NOAA-GOES17). They used the Python library s3fs (File System in User Space).
    - They used the Satpy Python library (a standard tool for processing meteorological satellite data) to read the raw, native GOES files directly from the mounted AWS directory.
    - Using Satpy, they converted the GOES radiances to Brightness Temperature using Planck's Law
    - They converted GOES projection into standard WGS84 (EPSG:4326) latitude and longitude coordinates to align with VIIRS